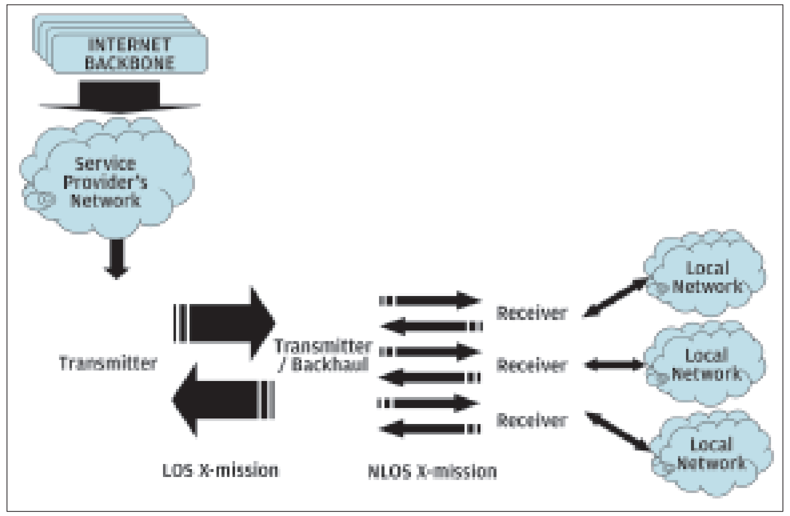


firms globally to shift telephony to VoIP, enhancing the importance of WiMAX. The shift to IP-based network enables much easier and cheaper network monitoring and management with standard tools for telecom operators.

- **Flatter and simpler topology**: a data network by design, WiMAX has a more simple topology compared to cellular networks which have undergone layer additions to accommodate data transmission capabilities. WiMAX also requires less time and fewer hardware, investment and human resource to set up than traditional cellular/Wi-Fi infrastructures, resulting in lower service costs for end users. WiMAX also provides the capability to scale up and down according to the application requirements.
- **Adaptability to a wide range of applications:** with its IP-based architecture, WiMAX supports a wide range of communication and data transfer applications. WiMAX also provides the possibility of transforming the way a large number of services such as internet, telephony, cable TV, etc. are currently provided to consumers.
- **Inexpensive interface**: the costs involved in migrating existing hardware to WiMAX are low, owing to the low prices of the required chipsets and interfacing devices.

How does WiMAX compare with WiFi?

Even though a large number of comparisons are made between WiMAX and WiFi, the similarities are limited when compared to the performance differences between these two technologies. WiMAX provides a wide range of advantages over Wi-Fi:

- **Wider reach**: Wi-Fi is short range (30-100 metres) compared to WiMAX, which provides a coverage radius of about 50 kms. This leads to Wi-Fi being limited to providing access only within a shop or a floor of a building, while WiMAX in comparison can cover much wider spaces.
- **Optimal bandwidth utilization**: advanced multiplexing provides better bandwidth utilization in WiMAX compared to WiFi.
- **Higher speeds**: Wi-Fi hotspots are typically backhauled over ADSL. Due to this, Wi-Fi access has poor upload speeds between the router and the internet compared to WiMAX. Additionally, no traffic prioritization leads to farther systems being made to wait longer than the ones close by in a Wi-Fi network. Proper scheduling in WiMAX standards takes care of this issue.
- **Limited hardware and manpower requirement**: to service an area covered by WiMAX using Wi-Fi networks would need a large number of hotspots to be created through routers. This necessitates a much higher manpower and hardware requirement for Wi-Fi compared to WiMAX which only requires a tower to be installed.
- **Noise reduction**: with stronger encryption/modulation, WiMAX performs better in noisy conditions.
- **Security**: with enhanced encryption, WiMAX offers better security than Wi-Fi networks which have been notorious for security breaches.

What are the potential applications of WiMAX?

The bandwidth, reach and IP-based topology of WiMAX make it suitable for a large number of potential applications. Some of the possible applications of WiMAX which stand to transform the way the whole information and communication business operates globally are:

- **Shifting telephony to VoIP**: globally, fixed-line telecom players are losing market share to mobile operators. WiMAX provides an opportunity for these players to transform their cable-based systems to IP-based VoIP system and provide additional high-speed data services.
- **Last mile broadband connectivity**: WiMAX is the ideal technology for providing a wireless alternative to cable and DSL for last mile broadband access to direct consumers.
- **Connecting remote locations**: remote locations with limited telecom infrastructure and sparse populations can be financially infeasible when connected through physical wire-based infrastructure. WiMAX provides a viable option to provide connectivity in these areas
- **Connecting Wi-FI hotspots**: WiMAX can be used to connect various Wi-Fi hotspots with each other and to other parts of the internet.

- **High speed data and telecom services**: WiMAX specifications can be used by telecom operators to design high speed communication and data services for consumers.
- **Providing Nomadic connectivity**: Mobile WiMAX would make it possible to provide good connectivity to communication devices in motion.
- **Connectivity systems for disaster situations:** during disasters and calamities physical infrastructure can become damaged, limiting connectivity in critical situations. WiMAX connectivity can be vital in such conditions.
- **Physical infrastructure substitution**: high speed data backhaul through WiMAX can lead to the transformation of many data intensive services by the substitution of the physical hardware by wireless connectivity. For example, Internet Protocol Television (IPTV) can substitute traditional cable-based TV, backhaul for Wi-Fi hotspots can shift from DSL, evolution of mobile data TV, emergency response services, wireless backhaul as substitute for fibre-optic cable, etc.

What are the major concerns about WiMAX?

Despite the mentioned advantages and the business potential, there are several potential issues and roadblocks for a WiMAX rollout in India:

- **Spectrum Allocations Issues:** globally, the roll-out of WiMAX in many nations has been delayed due to the required frequency bands being occupied for other applications. The situation was similar in India with a substantial portion of the required frequency being used by the military. With the recently planned auctions for WiMAX the roadblocks seem to be lifting.
- **Market downturn**: due to the recent economic downturn, firms may find it difficult to pull in financial resources to make the necessary investments for WiMAX rollout.
- **Other technologies**: availability of existing technologies and required investments may deter many firms from making the shift to WiMAX. Additionally, the long delayed 3G would also compete for market share in India, possibly leading to restricted growth for WiMAX.
- **Performance Issues:** In addition, some of the frequencies utilized by WiMAX are subject to interference from rainfade. The unlicensed WiMAX frequencies are subject to RF interference from competing technologies and competing WiMAX networks.

What's ahead for WiMAX?

Keeping in mind the technical superiority of WiMAX over other wireless networking technologies, the possibilities are rife that WiMAX would be used to develop higher generation data and communication technologies like 4G. For instance, the US based telecom firm Qualcomm has been pushing a standard for 4G known as 802.20, based on OFDMA, a multiplexing standard developed for WiMAX.

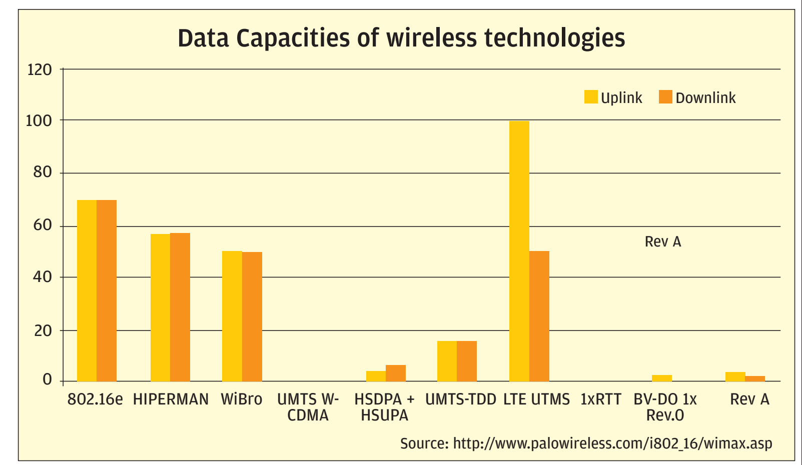
Further new technologies like 802.20 Mobile-Fi are expected to emerge challenging existing standards and providing better performance in terms of bandwidth, speed, reduced latency, encryption (noise reduction & security), et al.

3G vs. WiMAX

Comparison of existing wireless communication technologies

The communications space has seen an array of wireless technologies, each bringing in a new set of characteristics and raising performance standards. 3G, WiMAX, Wi-Fi and Bluetooth, are the prominent ones.

In terms of data transmission capabilities, WiMAX leaves all the currently available wireless technologies far behind. LTE, a technology being backed by Ericsson, is similar to WiMAX and is expected to surpass WiMAX in terms of performance in future.



Comparison of 3G and WiMAX

Both 3G and WiMAX have been touted as the next big hopes for the communications sector and particularly telecommunications technology. Globally, 3G has been present in the market for a period of time and has acquired a market share, while WiMAX is still a very recent phenomenon which has had limited deployments across the globe. A comparative analysis of both follows:

- **Existing infrastructure**: in many nations large telecom players have invested heavily in 3G technology in the last few years, creating